PROBLEM FORMULATION

Problem Formulation: Multi-Agent RL for Traffic-Signal Coordination

Four cooperating agents, one per intersection of a 2<2 SUMO grid, learn signal timing that cuts average waiting time versus a fixed-time baseline.

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Formulation. Each intersection is an agent in a cooperative, partially observable Markov game (Dec-POMDP): it sees only its own approaches and chooses its own phase, but all agents share one goal.

Environment

SUMO 2<2 grid, 4 traffic-light junctions, 3,600 one-second simulation steps per episode, stochastic vehicle demand with peak and off-peak periods

Reward

PER AGENT

Negative change in accumulated waiting time at its junction between decisions, so reducing queues earns positive reward; a shared network-wide term encourages coordination

State

PER AGENT

Queue length and cumulative waiting time on each incoming lane, current phase (one-hot), and time since the last phase change

Action

PER AGENT

Discrete: select the next green phase, subject to a minimum green time and a fixed yellow transition

Transition

SUMO car-following and signal logic; the next state depends on all four agents' joint actions

Objective

Maximise expected discounted return, which minimises average vehicle waiting time across the network

Algorithm and training

Deep Q-Learning (DQN) with experience replay and a target network, with one policy shared across the four intersections and an agent ID or local observation to tell them apart. Exploration uses epsilon-greedy with decay. Episode 1 (untrained) serves as the learning-curve starting point.

Evaluation

Six traffic metrics are tracked per simulation step: average wait time, stopped cars, average speed, cumulative delay, queue density and accumulated wait. The trained policy is compared with an untrained agent (episode 1) and with a fixed-time signal baseline on an unseen random seed, which tests for overfitting.